Federica Costantino, PhD

Contact

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Short Bio

I enjoy using a chemist's intuition to tackle problems, and I am passionate about the scientific investigation of materials. I had more than 10 years of research experience in diverse lab environments, which gave me the possibility to work on cross-functional and international teams and handle a variety of experimental setups and instruments; meantime the challenge to manage the experimental data focusing deeply on the details and organization of the work and research projects which allowed me to growth in professional development. Strongly consciousness the importance of the environment, and our role to preserve it, and for this, my motto is "leave the world better than we found it"

Work Experience

Nov 2022- to date – Technical Projected Manager, Water Area at Utilitalia - Working on different topics focused on the wastewater treatment processes and their related technology in order to improve better procedures to reduce pollution in water, understand the role of emerging pollutants. Working on circular economy aspect and the reduction of GHG emissions.

Jan 2022, October 2022- **Project Manager** at **RINA Consulting SpA - Centro Sviluppo Materiali** (Rome). Drafting and coordination of Horizon Europe proposals and national PNRR instruments; focusing on the decarbonization of energy-intensive industries and circular economy in the Steel sector.

Jan 2020- Dec 2021 - **Research Scientist** at the Faculty of Chemistry and Biochemistry at **the University of Notre Dame (USA)** - Development of catalytic membranes for the separation and generation of Hydrogen through photocatalytic processes and electrocatalysis for renewable energy demand in the US.

2017-2020– PhD Fellow Student at **Istituto Italiano di Tecnologia (IIT)** in the **Smart Materials** group (resp. Dr. Athanassia Athannassiou and Dr. Despina Fragouli) - Development of polymeric nanocomposites for water treatment applications by photocatalysis. The polymeric membrane is produced by electrospinning techniques using different matrix

2016-2017- **Material Chemist fellow** at **the CNR-ICCOM Institute** (Pisa) – Development and realization of polymeric materials with Phosphorene for opto-electronic applications.

2012-2013 - Chemist at A.S.P Palermo, chemical analysis and report on water and food for human trade.

Competence

•Scientific knowledge about plastic materials, surface treatment and engineering •Knowledge of pollutants and their environmental impact •Wastewater treatment process • Chemical analysis and synthesis of polymeric material •FT-IR and ATR analysis ì • Data analysis • Data analysis • Industrial process • MATLAB
• Micrococopy (SEM-TEM-Optical)• Mechanical analysis on polymeric materials (DMA) •Excellent knowledge of Microsoft Office •Project management •Scientific writing • Public speaking

Education

<u>2022</u> - Double PhD Degree in Chemistry Science at University of Notre Dame (USA) and the Catholic University of the Sacred Heart.
<u>2017</u> - Master's Degree in **Industrial Chemistry** at the University of Pisa (with a score of 110/110)
<u>2014</u> - Bachelor's Degree in **Chemical Sciences** at the University of Palermo

Skills

• Excellent analytical skills in managing complex processes/projects. • Coordination and planning of strategic actions to support the consortium to achieve the realization of the project. •Meet the deadline •Attention to detail during the drafting of research projects with a good analytical method. • Result and quality orientation• Attitude to Team working. • Manage stressful and complex situations • Circular economy aspect • Team working

Languages

• Italian • English (Fluent)

Hobbies

• Travel • Photography •Thrillers book

• Merbership Of Società Chimica Italiana • Finalist at Chimicapisce (Chemical Divulgation Contest)

List of Publications

1. **Costantino F..** Advanced Polymeric Nanocomposites for photocatalytic degradation of Pollutants in Water (2022)
2. **Costantino, F.** & Kamat, P. V. (2021). Do Sacrificial Donors Donate H2 in Photocatalysis?", ACS Energy Letters
3. **Costantino, F**., Gavioli, L., & Kamat, P. V. (2021). A Bipolar CdS/Pd Photocatalytic Membrane for Selective Segregation of Reduction and Oxidation Processes. *ACS Physical Chemistry Au*.
4. Loo, S. L., Vásquez, L., Zahid, M., **Costantino, F**., Athanassiou, A., & Fragouli, D. (2021). 3D Photothermal Cryogels for Solar-Driven Desalination. *ACS Applied Materials & Interfaces*.
5. **Costantino, F**., Cavaliere, E., Gavioli, L., Carzino, R., Leoncino, L., Brescia, R., ... & Fragouli, D. (2021). Photocatalytic Activity of Cellulose Acetate Nanoceria/Pt Hybrid Mats Driven by Visible Light Irradiation. *Polymers*, *13*(6), 912.
6. **Costantino, F**., Armirotti, A., Carzino, R., Gavioli, L., Athanassiou, A., & Fragouli, D. (2020). In situ formation of SnO2 nanoparticles on cellulose acetate fibrous membranes for the photocatalytic degradation of organic dyes. *Journal of Photochemistry and Photobiology A: Chemistry*, *398*, 112599.
7. **Costantino, F**., Serrano-Ruiz, M., Peruzzini, M., & Heun, S. (2018). Hybrid nanocomposites of 2D black phosphorus nanosheets encapsulated in PMMA polymer material: new platforms for advanced device fabrication. *Nanotechnology*, *29*(29), 295601.
8. Passaglia, E., Cicogna, F., **Costantino, F**., Coiai, S., Legnaioli, S., Lorenzetti, G. (2018). Polymer-based black phosphorus (bP) hybrid materials by in situ radical polymerization: an effective tool to exfoliate bP and stabilize bP nanoflakes. *Chemistry of Materials*, *30*(6), 2036-2048.